Adaptive Fabric Surface Defect Detection and Localization using Classical Texture Descriptors

Prakruti Patel
Enrollment No.: AU2440046
School of Engineering and Applied Science
Ahmedabad University
Ahmedabad, Gujarat, India

Kushal Rajyaguru
Enrollment No.: AU2440140
School of Engineering and Applied Science
Ahmedabad University
Ahmedabad, Gujarat, India

Chintan Thacker
Enrollment No.: AU2440215
School of Engineering and Applied Science
Ahmedabad University
Ahmedabad, Gujarat, India

Parthiv Karangiya
Enrollment No.: AU2440016
School of Engineering and Applied Science
Ahmedabad University
Ahmedabad, Gujarat, India

***Abstract*—Fabric defect detection is challenging because small defects may occupy only a limited part of an image and may have intensity values similar to the surrounding fabric. Our project investigates a classical digital image-processing approach for detecting and localizing fabric defects using local intensity and texture information. Following the previous experiments with GLCM, LBP, and local anomaly scoring, we extended the implementation to compare multiple pixel-level anomaly maps. The updated pipeline includes dark contrast, bidirectional contrast, local texture deviation, gradient deviation, and directional Gabor texture responses. These features are combined into several alternative scoring configurations to investigate whether complementary intensity and texture information can represent different types of defects more effectively than a single cue. Each score map is robustly normalized, and detection thresholds are selected using sampled pixels from training images only. The pipeline also includes overlapping local neighborhoods, full-resolution mask generation, evaluation on held-out defective images, false-alarm analysis on defect-free fabric, and visualization of predicted masks. This report describes the implemented methodology and evaluation framework. The quantitative comparison and visual results will be incorporated after the complete Colab experiments have been executed and verified.**

***Index Terms*—Fabric defect detection, defect localization, digital image processing, texture analysis, GLCM, LBP, Gabor filters, anomaly detection, adaptive thresholding.**

I. Introduction

Fabric inspection is an important application of digital image processing because defects can reduce the quality and usability of textile products. Automated inspection is difficult because fabric images contain repetitive weave patterns, local brightness variations, and irregularities that may be difficult to distinguish from normal texture.

Our project, *Adaptive Fabric Surface Defect Detection and Localization using Classical Texture Descriptors*, investigates whether classical image-processing methods can identify abnormal fabric regions without relying on a deep learning model. The objective is not only to identify unusual patterns but also to produce a mask indicating where the suspected defect occurs.

In the earlier stages, we investigated GLCM and LBP features and developed a training-free texture-analysis pipeline on the TILDA database. We later moved to the AITEX Fabric Image Database because it provides defective images with corresponding ground-truth masks. This allowed us to evaluate predicted masks at the pixel level instead of relying only on visual inspection.

The previous experiments also exposed limitations in the baseline approach. A score based mainly on dark contrast may miss bright defects or defects that alter the weave without producing a strong intensity change. In addition, expanding a positive patch into a complete rectangular region may produce inaccurate defect boundaries, while normal fabric regions can generate false alarms.

During the current stage, we extended the implementation to investigate additional intensity and structural cues. The updated pipeline combines local contrast, texture statistics, gradients, and directional Gabor responses. It generates full-resolution anomaly maps and provides a consistent framework for comparing different scoring configurations using training and held-out data.

The main work completed in the implementation is summarized as follows:

- Extended the anomaly-scoring pipeline beyond dark contrast.
- Implemented bidirectional intensity contrast to represent both brighter and darker local anomalies.
- Added local texture-deviation, gradient-deviation, and multi-orientation Gabor response maps.
- Implemented robust normalization and alternative feature-fusion configurations.
- Added training-only threshold selection and separate held-out evaluation procedures.

- Added full-resolution predicted masks, visual comparisons, and exportable evaluation tables.

II. Methodology

A. Dataset and Evaluation Protocol

We continue to use the AITEX Fabric Image Database (AFID), which provides fabric images and pixel-level defect annotations [6]. The implementation reads the image-mask pairing manifest, retains valid image-mask pairs, and combines multiple masks associated with the same image when necessary.

The valid defective image-mask pairs are divided into training and held-out test subsets using a fixed random seed. The split is performed at the image level so that patches from the same image do not appear in both subsets. This is important because overlapping patches from one image could otherwise introduce information leakage between training and testing.

A separate split of defect-free fabric images is maintained for examining false-positive detections on normal fabric. The normal-image evaluation is kept separate from the defective-image mask evaluation because defect-free images do not contain positive defect annotations.

The training images are used for threshold selection. The held-out defective images and held-out defect-free images are reserved for evaluation after the scoring configurations and thresholds have been selected.

B. Image Preprocessing

Each fabric image is loaded as a grayscale image and converted to a floating-point intensity representation for filtering and feature calculation. The original image dimensions are preserved throughout the processing pipeline.

A local background estimate is obtained by smoothing the image with a Gaussian filter. This provides a representation of the slowly varying surrounding intensity against which local pixels can be compared. The preprocessing is applied consistently before the anomaly maps are calculated.

Unlike a method that assigns a single label to an entire image, the updated implementation computes spatially varying score maps. Every score map has the same height and width as the input image, allowing the final prediction to be generated at the pixel level.

C. Bidirectional Local Contrast

The initial dark-contrast method identifies locations that are darker than their local background. If $I(x, y)$ denotes the input intensity and $B(x, y)$ denotes its smoothed local background estimate, the dark-contrast score is

$$D(x, y) = \max\left(B(x, y) - I(x, y), 0\right). \tag{1}$$

This score responds primarily to dark local anomalies. To avoid making the complete system dependent on this assumption, the updated pipeline also implements bidirectional contrast:

$$C(x, y) = \left|I(x, y) - B(x, y)\right|. \tag{2}$$

The absolute difference responds to intensity deviations in either direction. Therefore, it can represent locations that are brighter or darker than their surroundings.

The bidirectional contrast map is treated as a separate candidate as well as an input to score fusion. This makes it possible to compare the effect of two-sided intensity information with the dark-only baseline.

D. Local Texture Deviation

Fabric defects may disrupt the repetitive structure of the weave even when their intensity is not substantially different from that of the background. To capture such variations, the implementation calculates local intensity statistics using a sliding neighborhood.

For a neighborhood $\mathcal{N}_{x,y}$, the local mean and local standard deviation can be expressed as

$$\mu(x, y) = \frac{1}{|\mathcal{N}_{x,y}|} \sum_{(u,v)\in\mathcal{N}_{x,y}} I(u, v), \tag{3}$$

$$\sigma(x, y) = \sqrt{\frac{1}{|\mathcal{N}_{x,y}|} \sum_{(u,v)\in\mathcal{N}_{x,y}} \left(I(u, v) - \mu(x, y)\right)^2}. \tag{4}$$

The local standard-deviation map represents changes in local intensity variation. The implementation compares this map with a smoothed version of itself to obtain a texture-deviation response:

$$T(x, y) = \left|\sigma(x, y) - G_\rho * \sigma(x, y)\right|, \tag{5}$$

where G_ρ is a Gaussian smoothing kernel and $*$ denotes convolution.

This cue is intended to identify local changes in texture strength that may not be represented adequately by intensity contrast alone. It complements the texture descriptors investigated earlier in the project, including GLCM and LBP [1]–[3].

E. Gradient Deviation

Defect boundaries can change local edges and intensity transitions. The updated method therefore includes a gradient-based feature.

The horizontal and vertical image gradients are calculated using Sobel operators:

$$I_x(x, y) = \frac{\partial I}{\partial x}, \qquad I_y(x, y) = \frac{\partial I}{\partial y}. \tag{6}$$

The gradient magnitude is

$$M(x, y) = \sqrt{I_x(x, y)^2 + I_y(x, y)^2}. \tag{7}$$

A local reference is obtained by smoothing the gradient-magnitude map. The gradient-deviation score is then calculated as

$$E(x, y) = \left|M(x, y) - G_\rho * M(x, y)\right|. \tag{8}$$

This feature responds to local differences in edge strength and may help identify narrow lines, broken weave patterns, and other structural irregularities. It is evaluated independently and as part of combined score maps.

F. Directional Gabor Texture Analysis

The major addition to the previous pipeline is a Gabor filter bank. Woven fabric contains repeated directional patterns, so a defect may be represented by a disturbance in the texture response at a particular orientation.

Gabor filters provide a way to analyze local spatial frequency and orientation information. The implementation uses filters with different orientations to avoid relying on only one direction. Before applying the filters, a high-frequency image component is obtained by subtracting a smoothed version of the original image:

$$I_h(x, y) = I(x, y) - G_\eta * I(x, y). \tag{9}$$

Let g_θ denote a Gabor kernel at orientation θ. Its response magnitude is

$$R_\theta(x, y) = |I_h * g_\theta| . \tag{10}$$

For each orientation, the local response is compared with a smoothed reference:

$$A_\theta(x, y) = |R_\theta(x, y) - G_\rho * R_\theta(x, y)| . \tag{11}$$

The final Gabor-deviation map is obtained by retaining the strongest response across the filter orientations:

$$A_G(x, y) = \max_\theta A_\theta(x, y). \tag{12}$$

This allows the detector to respond to local deviations in directional texture patterns. Gabor filtering has also been investigated in fabric defect detection literature, including work comparing GLCM and Gabor features [5].

The Gabor response is used both as an individual anomaly map and as an input to the fusion configurations described below.

G. Robust Score Normalization

The different anomaly maps represent different image properties and may have different numerical ranges. Directly adding them without normalization could allow one feature to dominate because of its scale rather than its usefulness.

The implementation therefore applies robust normalization to each map. For a raw anomaly map S, let m_S be its median and let $\mathrm{MAD}(S)$ denote its median absolute deviation. A robust scale is obtained using the median absolute deviation and upper quantiles of the map.

The normalized positive anomaly response is calculated in the form

$$Z_S(x, y) = \mathrm{clip}\left(\frac{S(x, y) - m_S}{s_S + \epsilon}, 0,\ 12 \right), \tag{13}$$

where s_S is the robust scale estimate and ϵ is a small constant for numerical stability.

Robust normalization reduces the influence of extreme responses and places the candidate feature maps on more comparable scales. The normalization is applied consistently before constructing the fusion scores.

H. Feature Fusion Configurations

A single anomaly cue may be insufficient because fabric defects can change intensity, local texture, or directional structure in different ways. The implementation therefore creates several alternative fusion maps.

The texture-gradient fusion combines local texture deviation and gradient deviation:

$$F_{TG} = \mathcal{N}\left(0.60Z_T + 0.40Z_E\right), \tag{14}$$

where $\mathcal{N}(\cdot)$ denotes robust normalization.

A texture-Gabor fusion combines statistical texture variation and directional texture responses:

$$F_{TG} = \mathcal{N}\left(0.55Z_T + 0.45Z_G\right). \tag{15}$$

To investigate a configuration that does not rely on dark contrast, the non-dark fusion combines bidirectional contrast, texture deviation, gradient deviation, and Gabor texture:

$$F_{ND} = \mathcal{N}\left(0.25Z_C + 0.20Z_T + 0.10Z_E + 0.45Z_G\right). \tag{16}$$

The implementation also includes an all-cue fusion that incorporates both dark and bidirectional contrast along with texture and structural features:

$$F_{\text{all}} = \mathcal{N}\left(0.15Z_D + 0.20Z_C + 0.20Z_T + 0.10Z_E + 0.35Z_G\right). \tag{17}$$

Finally, a smoothed version of the all-cue map is generated to examine whether mild spatial smoothing can reduce isolated responses in the anomaly map.

These configurations are treated as separate candidates. They are not assumed to have equal performance, and their relative usefulness must be established using the same data protocol and evaluation metrics.

I. Threshold Selection Using Training Data

After feature extraction and fusion, the anomaly maps must be converted into binary defect masks. A single fixed threshold may not work equally well for every scoring method because their score distributions differ.

The implementation samples pixels from the training image-mask pairs and evaluates candidate thresholds derived from score percentiles. Each sampled pixel inherits its label from the corresponding ground-truth mask. For each method, candidate thresholds are compared using training-sample precision, recall, F1-score, and IoU.

TABLE I
IMPLEMENTED ANOMALY-SCORING CONFIGURATIONS

Configuration	Information used
Dark contrast	Darker-than-background intensity
Bidirectional contrast	Brighter or darker local intensity
Texture deviation	Local intensity variation
Gradient deviation	Local edge-strength changes
Gabor texture	Directional texture responses
Texture-gradient fusion	Texture and gradient cues
Texture-Gabor fusion	Texture statistics and directional responses
Non-dark fusion	Bidirectional contrast, texture, gradient, and Gabor
All-cue fusion	Dark contrast and complementary cues
Smoothed fusion	Spatially smoothed all-cue score

The selected threshold for each method is the one that provides the highest training-sample F1-score under the implemented selection rule. The selected threshold is then frozen for held-out evaluation.

The held-out test images are not used to choose feature weights or thresholds. This prevents test-set feedback from influencing the reported evaluation.

J. Full-Resolution Mask Generation

The previous patch-based approach expanded a positive patch into a rectangular region. This can create coarse masks if a defect occupies only a small portion of the patch.

The extended pipeline instead generates pixel-level score maps directly. For a method with score map $F(x, y)$ and selected threshold τ, the predicted binary mask is

$$\widehat{M}(x, y) = \begin{cases} 1, & F(x, y) \geq \tau, \\ 0, & F(x, y) < \tau. \end{cases} \tag{18}$$

The threshold is applied at the original image resolution, allowing the predicted mask to follow local score changes instead of forcing every detection to occupy a complete patch. The resulting mask is stored for visual comparison with the ground-truth annotation.

K. Evaluation Framework

The evaluation framework reports pooled pixel-level metrics across the held-out defective images and mean per-image metrics separately. Precision measures the fraction of predicted defect pixels that are correct, recall measures the fraction of ground-truth defect pixels recovered, F1-score balances precision and recall, and intersection over union measures overlap between the predicted and ground-truth masks.

False-positive rates are also calculated on held-out defect-free images. This provides an additional way to assess whether a method detects defects while avoiding excessive alarms on normal fabric.

The implementation saves per-image metrics, pooled metrics, mean per-image metrics, threshold-selection tables, and prediction masks. It also generates visual comparisons between the original fabric image, the ground-truth mask, and the predictions from different scoring configurations.

This common evaluation procedure makes the methods comparable without selecting a different test protocol for each candidate.

III. IMPLEMENTATION PROGRESS

During this stage, the Colab implementation was extended from a single primary contrast-based detector into a comparative classical-image-processing pipeline.

First, the image-loading and evaluation code was organized around the AITEX image-mask manifest. The implementation validates image and mask dimensions and combines the masks associated with each valid defective image. A reproducible image-level train/test split is used so that the training and held-out subsets remain separate.

Second, the feature-extraction stage was expanded. In addition to dark contrast, the pipeline calculates bidirectional local contrast, local texture deviation, gradient deviation, and a multi-orientation Gabor texture response. Each feature is calculated as a spatial map rather than a single score for an entire image.

Third, the individual maps are robustly normalized and combined into multiple fusion configurations. This allows the evaluation to examine whether local texture and directional structure provide complementary information beyond intensity contrast. A separate non-dark fusion configuration is included to study a detector that does not depend on the original dark-only cue.

Fourth, the threshold-selection process was implemented using pixels sampled from training images. Candidate score percentiles are compared using training-sample metrics, and the selected threshold for each method is stored for subsequent evaluation.

Finally, the notebook includes a held-out evaluation workflow. The defective test images are evaluated against their ground-truth masks, while defect-free test images are used to examine false-positive detections. The code also prepares visual mask comparisons and exports the evaluation tables and generated masks for later analysis.

The implementation therefore provides a consistent framework for studying several candidate feature combinations. The final numerical comparison and visual evidence are intentionally not included in this report because the complete Colab execution and verification are still required.

IV. DISCUSSION

The extension of the pipeline addresses two limitations identified in the earlier experiments: the restricted response of dark-only contrast and the coarse localization produced by block-shaped masks.

Bidirectional contrast allows the score map to respond to deviations on either side of the local intensity reference. Texture deviation and gradient deviation add information about local variation and edge structure. Directional Gabor responses

provide another representation of changes in the repeated weave pattern.

The fusion configurations combine these complementary representations in different ways. In particular, the non-dark configuration allows the contribution of texture and structural features to be investigated without including the dark-contrast map. The all-cue configuration retains the dark-contrast baseline while incorporating the other features.

A further component of the implementation is the separation between threshold calibration and evaluation. Candidate thresholds are selected from training data, while held-out defective images are reserved for measuring localization performance. Held-out defect-free images are used to examine false-positive behavior.

The use of full-resolution score maps also changes how defect regions are generated. Instead of assigning every pixel in a positive patch to the defect class, the score is thresholded spatially. This removes the requirement that every detection must have a block-shaped boundary, although the usefulness of this approach must still be established through the completed experiments.

The added methods are candidates rather than guaranteed improvements. Gabor responses, additional feature fusion, and pixel-level thresholding may still produce false positives on normal weave patterns. Therefore, the numerical evaluation and visual inspection are necessary before deciding which configuration is suitable for the final model.

V. Conclusion

During Week 6, we extended the classical fabric defect detection implementation to include bidirectional contrast, local texture deviation, gradient deviation, directional Gabor filtering, and alternative score-fusion configurations. The updated pipeline produces full-resolution anomaly maps, normalizes the component features, and selects thresholds from training data rather than using held-out test results.

The implementation also provides a consistent evaluation framework for defective and defect-free images, generation of binary masks, visual comparisons, and export of metrics for each candidate method. These additions make it possible to compare intensity-based and texture-based approaches under the same evaluation protocol.

The new methodology is designed to address the limitations of dark-only scoring and coarse patch localization. However, no claim of improved accuracy is made at this stage. The final comparison will be based on the executed Colab experiments and will be added with the corresponding tables and visualizations after the outputs have been verified.

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